How Firm ESG Signaling and Third-Party Certifier Credibility Shape Investment Decisions in Markets Prone to Greenwashing

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Abstract

This paper studies how third-party certifier credibility affects investment decisions in an ESG market where firms can either make genuine sustainability investments or greenwash. I use an agent-based evolutionary simulation in Python with 500 firms, one certifier, and investors over 100 periods. Genuine ESG firms draw quality from a $\text{Beta}(8, 2)$ distribution and pay a higher implementation cost, while greenwashing firms draw from $\text{Beta}(2, 8)$ and pay a lower signaling cost. Investors allocate capital through a logistic decision rule that combines financial return, certifier reputation, investor trust, firm quality, and a modest warm-glow preference for ESG-labeled assets. I compare a high-credibility certifier ($a = 0.90$) with a low-credibility certifier ($a = 0.60$). In the seeded run reported here, both scenarios move from the initial 50–50 firm split toward separating-like outcomes, but high certification credibility produces cleaner capital allocation. Over the final 30 periods, the high-credibility scenario has an average greenwashing share of 14.8%, a greenwashing investment success rate of 8.7%, and an investor decision success rate of 98.3%. The low-credibility scenario has an average greenwashing share of 15.0%, a greenwashing investment success rate of 13.0%, and an investor decision success rate of 95.5%. These results are not market forecasts. Within this simplified computational setting, they show that certifier credibility changes the precision of ESG market separation even when greenwashing does not disappear completely.

Keywords: ESG signaling, greenwashing, certifier credibility, evolutionary simulation, asymmetric information, bounded rationality

1 Introduction

1.1 The asymmetric information problem in ESG markets

Environmental, social, and governance (ESG) investing depends on an information problem. Firms know more about their actual sustainability practices than outside investors do. Investors usually see disclosures, ratings, certifications, and marketing claims, then decide whether those signals are credible enough to guide capital allocation. This creates room for greenwashing, where firms imitate sustainability signals without paying the full cost of genuine sustainability investments.

Greenwashing matters because it weakens the meaning of ESG signals. If investors cannot distinguish genuine firms from firms that mainly advertise sustainability, the ESG label becomes less informative. Genuine firms then face a harder task because they must compete with cheaper imitation. This is close to the adverse selection problem described in signaling theory, where a signal loses value when low-quality participants can mimic it.

The damage is not limited to individual mistakes. When greenwashing spreads, investors may discount ESG claims generally, not only the claims of firms later caught misleading the market. That makes capital allocation less precise. In the extreme, investors may either overfund weak ESG projects because the label feels credible or underfund genuine firms because the label feels contaminated.

1.2 Greenwashing as a verification problem

Recent regulatory and academic work suggests that greenwashing is not a marginal concern. The European Banking Authority documents a 21.1% global increase in alleged greenwashing cases between 2022 and 2023 (European Banking Authority, 2024). That number is not used here as a direct model parameter, but it motivates the central concern that verification infrastructure may not be keeping pace with the growth of ESG-labeled products.

Media and public scrutiny also matter. Ben Mahjoub (2024) argues that media coverage can either expose greenwashing or help reinforce it by repeating weak claims. In market terms, media attention acts as a secondary signal. It can raise the expected cost of deception, but it can also make investors skeptical of the entire ESG category if scandals dominate what they observe.

Third-party certification is one possible response. A credible certifier can turn private information about firm quality into a more useful public signal. A weak certifier can do the opposite by giving greenwashing firms the same label as genuine firms, which reduces the value of certification itself. The question in this paper is therefore not simply whether ESG certification matters, but how certification accuracy changes simulated market dynamics when firms, certifiers, and investors adjust over time.

1.3 Research objective and assumptions

This paper builds an agent-based evolutionary simulation to isolate that mechanism. The model is intentionally stylized and does not estimate real ESG fund flows, real certification error rates, or real investor preferences. Instead, it asks whether changing certifier accuracy changes the separation between genuine and greenwashing firms under fixed assumptions.

The model treats firms as boundedly rational. They do not solve an infinite-horizon optimization problem. Instead, they compare recent realized returns and sometimes imitate the strategy that has performed better. Certifier reputation updates from observed certification performance. Investor trust updates from realized investment outcomes. These are simplifications, but they allow the simulation to track feedback that is difficult to observe directly in real markets.

1.4 Definitions

ESG signaling refers to the process through which firms communicate sustainability practices to investors. In the model, the signal includes both the firm's underlying ESG quality q_i and the certifier's decision.

Greenwashing refers to a low-cost imitation strategy in which a firm attempts to appear sustainable without making the same level of genuine sustainability investment.

Certifier credibility refers to the reliability of the third-party verification process. In the model, the certifier's underlying accuracy is represented by a, while its public reputation evolves as R_t.

Investor trust refers to the willingness of investors to treat certified ESG signals as informative. In the model, investor trust evolves as T_t based on realized investment outcomes.

1.5 Research question

The central research question asks:

> How does the interaction between firm ESG signaling and third-party certifier credibility shape investment decisions in a market prone to greenwashing?

I answer this question by comparing two simulated markets. In one, the certifier has high accuracy ($a = 0.90$). In the other, the certifier has lower accuracy ($a = 0.60$). The comparison focuses on greenwashing persistence, greenwashing investment success, investor decision success, certifier reputation, and investor trust.

2 Literature Review

2.1 Signaling theory and ESG markets

Spence's signaling model provides the theoretical starting point (Spence, 1973). In markets with hidden quality, signals are useful when they are costly enough that low-quality agents cannot easily imitate high-quality agents. Applied to ESG markets, genuine sustainability investment should be a costly signal because it requires operational changes, compliance costs, and long-term commitment. Greenwashing is the cheaper imitation strategy.

This logic is also connected to reputation. Recent work on sustainable finance and corporate reputation treats sustainability claims as signals that affect outside perceptions of firm quality (Arhinful et al., 2025). The problem is that a signal only works if the market believes it is tied to real cost or credible verification. If investors only see firm claims, the market can move toward pooling, where genuine and deceptive firms issue similar ESG signals and investors cannot reliably infer type.

Certification can add a screening layer. A highly accurate certifier makes imitation harder because greenwashing firms are more likely to be rejected or later discovered, while a low-accuracy certifier weakens the screen and allows more deceptive firms to receive the same label as genuine firms.

2.2 Greenwashing and verification

Research on greenwashing emphasizes that deceptive ESG behavior can persist when enforcement is weak or claims are difficult to verify. Delmas and Burbano (2011) describe greenwashing as a corporate strategy shaped by external pressure, incentives, and weak monitoring. Dempere et al.

(2024) similarly frame greenwashing as a deceptive practice in which organizations devote more effort to appearing environmentally responsible than to reducing environmental impact.

The model simplifies this logic into two firm strategies. Genuine ESG firms pay a higher cost and tend to have higher quality. Greenwashing firms pay a lower cost and tend to have lower quality, but they can sometimes pass through certification. This simplification is useful because it makes the core tradeoff visible. Greenwashing survives when the expected reward from appearing sustainable exceeds the expected cost of rejection, detection, and penalty.

Investor and consumer skepticism is part of the same problem. Aji and Sutikno (2015), Nyilasy et al. (2012), and Andreoli and Minciotti (2023) all examine how perceived greenwashing can change trust, skepticism, and attention. These studies matter for the model because investor response is not automatic. Even when information exists, investors must decide how much weight to place on it.

2.3 Third-party assurance and certifier failure

Third-party assurance can improve the credibility of sustainability reporting. Pizzi et al. (2024) argue that external assurance can help restore trust in sustainability reporting, while Gipper et al. (2024) find that ESG assurance is associated with disclosure quality, ESG ratings, and institutional investor holdings. The important point for this paper is not that assurance always fixes greenwashing, but that assurance changes the information available to investors.

Certification systems can also fail, and Nygaard (2023) discusses cases in which certification has been accused of legitimizing questionable sustainability claims. That makes certifier accuracy and reputation central variables rather than background assumptions. In this simulation, the high-credibility and low-credibility scenarios are meant to capture that institutional difference in a simplified way.

2.4 Investor psychology and warm-glow demand

Investor decisions are not always purely financial, since some investors may receive non-financial satisfaction from holding ESG-labeled assets. Tatomir et al. (2023) discuss warm-glow investing, where investors are drawn toward ESG assets even when those assets may be vulnerable to greenwashing. In the simulation, this is represented by a modest warm-glow term in investor utility.

This behavioral term matters because it creates a demand-side vulnerability. If investors receive utility from the label itself, then greenwashing firms that obtain certification may attract capital even when their underlying ESG quality is low. Balaskas et al. (2025) connect this concern to ESG-labeled digital advertising and younger consumers. I do not model demographic differences directly, but this literature supports the broader assumption that ESG labels can carry psychological value beyond financial return.

2.5 Analytical gap

Much existing research emphasizes individual parts of the ESG information problem, including firm disclosure, greenwashing incentives, assurance quality, consumer skepticism, and investor preferences. This paper focuses on the feedback among those parts. Firm strategies change after observing recent returns. Certifier reputation changes after certification performance is revealed. Investor trust changes after investment outcomes are realized. A computational model is useful here because the mechanisms are linked and evolve together.

3 Methodology

3.1 Model overview

The model is an agent-based evolutionary simulation written in Python. It includes 500 firms and runs for 100 discrete periods. Each firm is assigned either genuine ESG implementation or greenwashing as its strategy. A certifier evaluates firms and decides whether to grant ESG certification. Investors then decide whether to allocate capital to certified firms.

The simulation compares two certification scenarios:

- **High-credibility certification** with certifier accuracy $a = 0.90$.
- **Low-credibility certification** with certifier accuracy $a = 0.60$.

The paper version of the code uses the same simulation structure as the earlier working version of the replication script, with one reproducibility adjustment. The two simulation calls explicitly pass `seed=42`. This makes the reported figures and tables deterministic. The earlier working version defines a seed argument in the class constructor but does not pass it in the two bottom-level simulation calls. Because of that, the public repository version should be run with the supplied seed to reproduce the manuscript numbers exactly.

3.2 Firm types and ESG quality

Each firm has an ESG quality score $q_i \in [0, 1]$. Genuine ESG firms draw quality from

$$q_i \sim \text{Beta}(8, 2), \tag{1}$$

while greenwashing firms draw quality from

$$q_i \sim \text{Beta}(2, 8). \tag{2}$$

These distributions make genuine firms higher quality on average while still allowing overlap. This reflects the fact that real sustainability quality is difficult to classify perfectly from outside.

Genuine ESG firms pay cost $c_G = 0.4$. Greenwashing firms pay cost $c_W = 0.1$. The base return is $R_f = 1.0$, and genuine ESG firms receive an ESG premium $\pi_{ESG} = 0.5$.

3.3 Certification process

The certifier evaluates each firm with accuracy a. A genuine ESG firm receives certification with probability a. A greenwashing firm is mistakenly certified with probability $1 - a$. The accuracy value is fixed within each scenario, while the certifier's reputation updates over time.

Certification performance in each period is the proportion of correct certification decisions. Certifier reputation then updates as

$$R_{t+1} = \text{clip}\left(R_t + \lambda(P_t - R_t),\, 0.1,\, 1.0\right), \tag{3}$$

where R_t is certifier reputation, P_t is certification performance in period t, and $\lambda = 0.1$ is the reputation adjustment rate. The clipping bounds prevent reputation from becoming unrealistically perfect or collapsing to zero.

3.4 Investor decision rule

For a certified firm i, investor utility is

$$U_I = r_i(T_t R_t) + \omega(0.5 + 0.5q_i) + 0.1q_i, \tag{4}$$

where r_i is the financial return, T_t is investor trust, R_t is certifier reputation, and $\omega = 0.15$ is warm-glow bias.

For genuine firms,

$$r_i = R_f + \pi_{ESG} - c_G. \tag{5}$$

For greenwashing firms,

$$r_i = R_f - c_W. \tag{6}$$

Investment probability follows a logistic response function:

$$P(\text{Invest}) = \frac{1}{1 + \exp[-k(U_I - \theta)]}, \tag{7}$$

where $k = 10$ is decision sensitivity and $\theta = 0.7$ is the skepticism threshold.

If investment occurs in a greenwashing firm, deception may be discovered with probability

$$P(\text{discovery}) = R_t \delta(1 - q_i), \tag{8}$$

where $\delta = 0.7$ is detection intensity. If discovered, the firm receives a penalty of 0.6.

Investor trust updates after each period. If investments occur,

$$T_{t+1} = 0.8T_t + 0.2S_t, \tag{9}$$

where S_t is the share of invested firms that were genuine. If no investments occur,

$$T_{t+1} = 0.95T_t. \tag{10}$$

3.5 Evolutionary strategy updating

Firms are boundedly rational rather than perfectly optimizing. This updating structure is motivated by evolutionary game theory, where agents adapt through imitation, selection, and bounded adjustment rather than solving a fully rational optimization problem (Alexander, 2022). They compare their recent returns to the recent returns of the alternative strategy using a five-period lookback window. With probability 0.3, a firm considers imitation. A switch occurs only if the alternative strategy has an average return at least 20% higher than the current strategy average and at least 10% higher than the firm's own realized return. Random mutation occurs with probability 0.05, allowing strategy changes even when imitation conditions are not met.

When a firm switches strategy, it receives a new ESG quality draw from the distribution associated with its new strategy. Firms that do not switch retain their existing quality score.

3.6 Simulation procedure

The replication script evaluates each period through the following sequence:

1. The current firm types are copied as a period-level snapshot.
2. The certifier assesses firms and issues ESG certifications.
3. Investors make investment decisions based on certified signals and utility calculations.
4. Returns are realized, and greenwashing detection occurs probabilistically.
5. Certifier reputation and investor trust update based on period outcomes.
6. Return history is updated, and firms evolve for the next period.
7. Period-level statistics are saved for analysis.

The main population and investment metrics use the pre-evolution firm-type snapshot for that period. This means that a period's greenwashing share and greenwashing investment success describe the market before the next strategy update takes effect.

3.7 Outcome metrics and threshold classification

The model tracks greenwashing proportion, greenwashing investment success, investor decision success, investment rate, average returns, certifier reputation, certifier performance, investor trust, and average ESG quality diagnostics.

Greenwashing investment success is the number of invested greenwashing firms divided by the number of greenwashing firms in that period's pre-evolution snapshot. **Investor decision success** is the share of invested firms that are genuine ESG firms under the same snapshot.

The code classifies final-window outcomes using thresholds. I describe these classifications as *separating-like*, *pooling-like*, or *mixed* rather than as formal game-theoretic equilibria, because the simulation does not analytically prove equilibrium existence or stability.

A separating-like outcome is defined as a final-window greenwashing proportion below 25% and greenwashing investment success below 35%. A pooling-like outcome is defined as a final-window greenwashing proportion above 55% and greenwashing investment success above 45%. Cases between those thresholds are classified as partial separating or mixed outcomes.

Table 1: Initial Conditions and Parameter Values

Parameter	Symbol	Value
Firm population	N	500
Time periods	T	100
Initial firm split	–	50% genuine, 50% greenwashing
Base financial return	R_f	1.0
ESG premium	π_{ESG}	0.5
Cost of genuine ESG	c_G	0.4
Cost of greenwashing	c_W	0.1
Greenwashing penalty	–	0.6
Detection intensity	δ	0.7
Warm-glow bias	ω	0.15
Skepticism threshold	θ	0.7
Decision sensitivity	k	10
Lookback window	–	5 periods
Mutation rate	–	0.05
Imitation strength	–	0.3
Initial certifier reputation	R_0	0.8
Initial investor trust	T_0	0.7
Certifier reputation gain	λ	0.1
High certifier accuracy	a_{high}	0.90
Low certifier accuracy	a_{low}	0.60

4 Results

Table 2 summarizes the seeded run used to generate the manuscript figures. Both scenarios satisfy the threshold criteria for separating-like outcomes. The high-credibility scenario, however, separates more cleanly in terms of investment outcomes.

Table 2: Summary of Main Simulation Outcomes, Seeded Code Run

Metric	High Credibility	Low Credibility
Outcome type	Separating-like	Separating-like
Final-window greenwashing share	14.8%	15.0%
Greenwashing investment success	8.7%	13.0%
Investor decision success	98.3%	95.5%
Final investor trust	0.986	0.954
Final certifier reputation	0.904	0.600
Recorded ESG quality gap	39.2 percentage points	40.8 percentage points

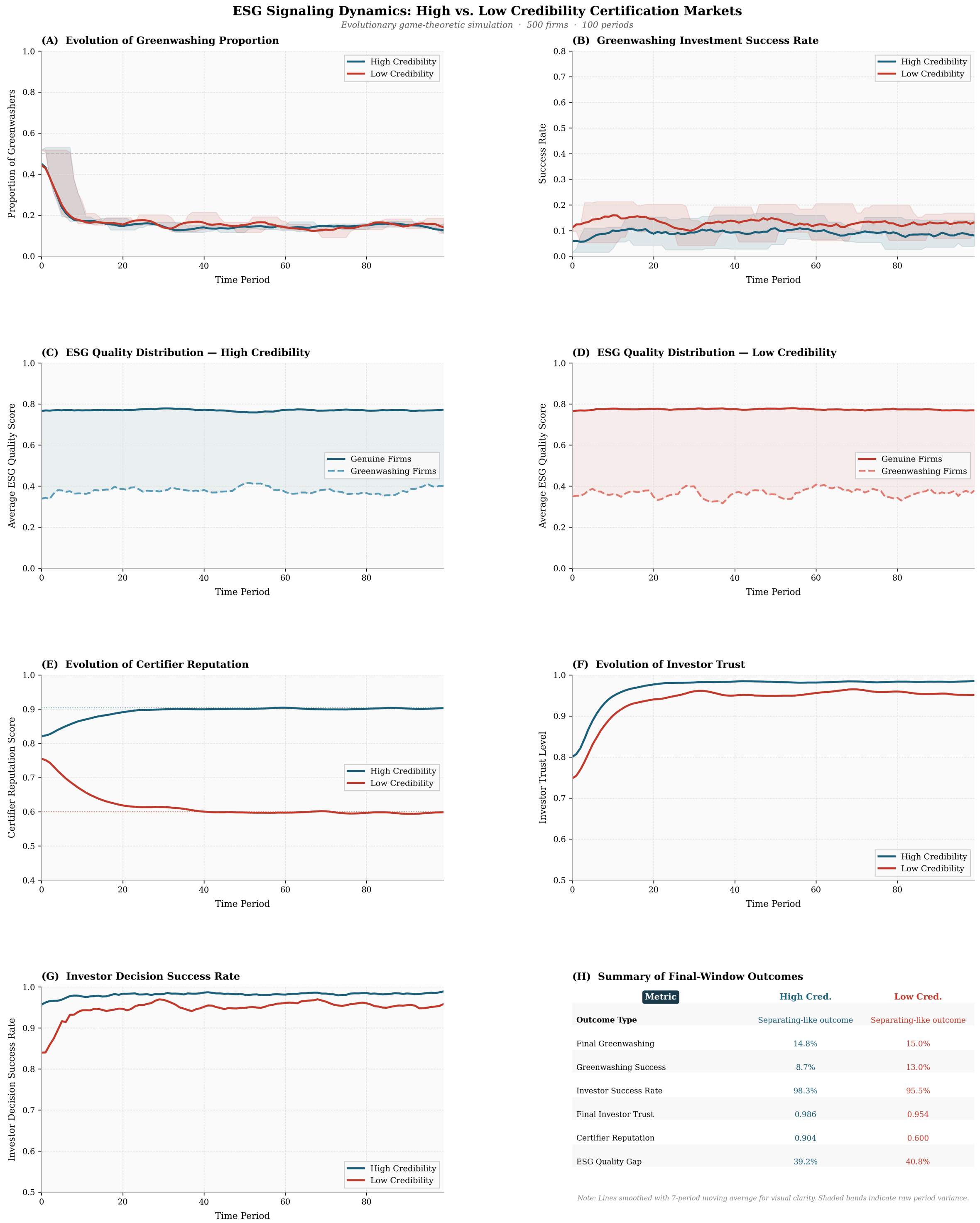

Figure 1: Main simulation dynamics for the high- and low-credibility certifier scenarios. Lines are smoothed for visual clarity, while the shaded bands show short-run period-level variation. The figure is generated from the Python simulation code used for the paper run.

4.1 Greenwashing proportion

Both scenarios begin with a 50–50 split between genuine and greenwashing firms. In both cases, greenwashing drops sharply during the early periods as firms respond to realized returns. In the seeded run, the average greenwashing share over the final 30 periods is 14.8% in the high-credibility scenario and 15.0% in the low-credibility scenario.

The lower-credibility certifier does not produce a much larger greenwashing population under these assumptions. The clearer difference is how often greenwashing firms successfully attract investment.

4.2 Greenwashing investment success

The high-credibility scenario produces a greenwashing investment success rate of 8.7%, compared with 13.0% in the low-credibility scenario. This is the strongest difference between the two markets because a lower-accuracy certifier mistakenly certifies more greenwashing firms, allowing more deceptive firms to enter the investor decision process.

Certifier credibility still matters even when both scenarios end in separating-like outcomes. The difference is not only how many greenwashing firms remain in the market, but how often those firms are able to capture capital.

4.3 Certifier reputation and investor trust

Certifier reputation ends at 0.904 in the high-credibility scenario and 0.600 in the low-credibility scenario. Investor trust follows a similar pattern, ending at 0.986 under high credibility and 0.954 under low credibility. These values show the feedback structure of the model. Accurate certification improves reputation and supports investor trust, while noisier certification produces more errors, lowers reputation, and reduces the usefulness of certification as a signal.

4.4 ESG quality diagnostics

The simulation assigns genuine and greenwashing firms to different quality distributions, centered near 0.8 and 0.2 respectively. In the recorded diagnostics from the seeded run, the final-window quality gap is about 39.2 percentage points under high credibility and 40.8 percentage points under low credibility. These diagnostics should be read as descriptive checks, not as independent empirical estimates. The main result of the paper does not rely on the precise quality-gap value. It relies on the comparison between certification regimes under the same model structure.

5 Discussion

The simulation suggests that certifier credibility affects the *quality* of ESG market separation rather than simply determining whether separation occurs. Both high- and low-credibility markets satisfy the threshold criteria for separating-like outcomes in the seeded run, but the high-credibility market produces lower greenwashing investment success and higher investor decision success.

The persistence of some greenwashing in both scenarios is not a contradiction. The mutation rate continually reintroduces greenwashing into the population, and certification and detection are probabilistic. Complete elimination is therefore unlikely in this finite evolutionary model, so the more realistic simulated outcome is a stable residual level of greenwashing.

One useful interpretation is that greenwashing proportion alone is not enough to evaluate ESG market health. In this run, both scenarios have relatively low greenwashing shares, but the low-credibility market still allows more greenwashing firms to receive investment. This means that a market can appear separated at the population level while still misallocating more capital through weaker certification.

The model also shows why certifier reputation matters as a dynamic variable. Reputation does more than summarize past performance because it also affects future investor decisions through the utility function. In the high-credibility scenario, reputation reinforces trust. In the low-credibility scenario, reputation falls toward the certifier's actual performance level. This feedback loop is central to the model's behavior.

6 Limitations and Future Research

This paper uses a stylized simulation, so the numerical results should not be read as real-world estimates. The model uses one certifier per scenario, two firm strategies, one representative investor decision rule, and fixed parameter values. Real ESG markets include competing rating agencies, heterogeneous investors, media scrutiny, regulatory enforcement, industry differences, and repeated firm-specific reputational effects.

Several parameter choices are exploratory. The mutation rate, penalty size, ESG premium, warm-glow coefficient, and skepticism threshold were selected to produce interpretable model behavior rather than calibrated from market data. That choice is reasonable for a mechanism-focused simulation, but it limits external validity.

A second limitation is the limited scenario space. The paper compares $a = 0.90$ and $a = 0.60$, but does not run a full sensitivity analysis across certifier accuracy values, penalties, ESG premia, or warm-glow levels. A stronger version of the study would run many Monte Carlo trials for each parameter setting and report confidence intervals.

A reproducibility issue in the earlier working version of the replication script is also worth stating directly. The simulation class supports a seed argument, but the earlier bottom-level calls did not pass one. The paper run corrects this by passing `seed=42` to both scenarios. For a public repository, the exact code used for the paper should be archived along with the generated output files.

Future work could extend the model in three directions. First, it could run a wider sensitivity analysis across certifier accuracy values, penalties, ESG premia, and warm-glow levels. Second, it could introduce heterogeneous investors with different levels of skepticism and ESG preference. Third, it could allow multiple certifiers to compete, which would make it possible to study certification shopping and reputation competition.

7 Conclusion

This paper built an agent-based evolutionary simulation to examine how ESG signaling works when firms can greenwash and investors rely on third-party certification. The main result is that certifier credibility changes the precision of market separation. In the seeded run reported here, both high- and low-credibility scenarios reach separating-like outcomes, but the high-credibility certifier produces lower greenwashing investment success and higher investor decision success.

The result should be interpreted carefully because the simulation does not prove that certifier credibility is the only determinant of ESG market efficiency, and it does not forecast real ESG investment flows. It shows that, within this parameterization, verification quality can affect whether investors convert ESG labels into accurate capital allocation. The broader implication is simple: ESG labels are only as informative as the institutions that verify them.

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Appendix A Replication Materials and Code Availability

The simulation was implemented in Python using NumPy, pandas, matplotlib, and SciPy. The exact code used for the manuscript run, along with generated figures and output files, is available in the project repository:

`https://github.com/f3z8/ESG-Signaling-Third-Party-Certifier-Credibility-and-Greenwashing`

The reported manuscript results use `seed = 42` for both the high-credibility and low-credibility certification scenarios. This makes the main tables and figures reproducible from the archived code.

Appendix B Additional Simulation Figures

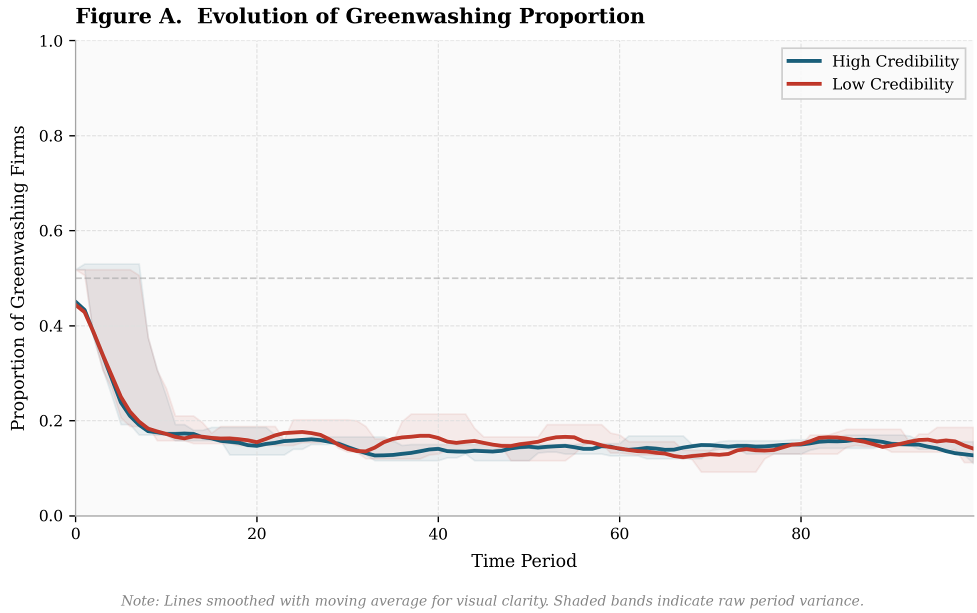

Figure 2: Greenwashing proportion over time in the high- and low-credibility certifier scenarios.

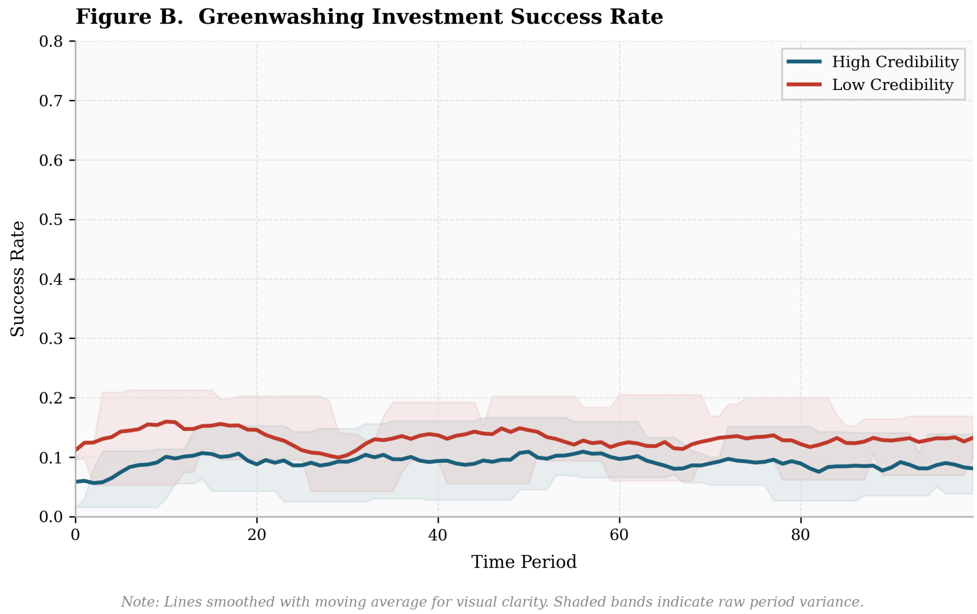

Figure 3: Greenwashing investment success rate over time.

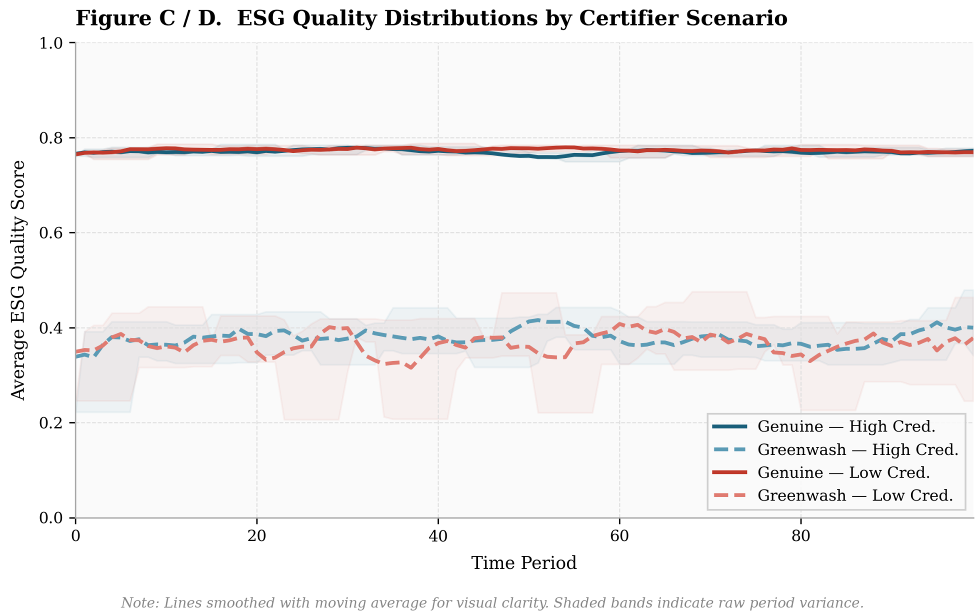

Figure 4: Average ESG quality diagnostics for genuine and greenwashing firms across both certifier scenarios.

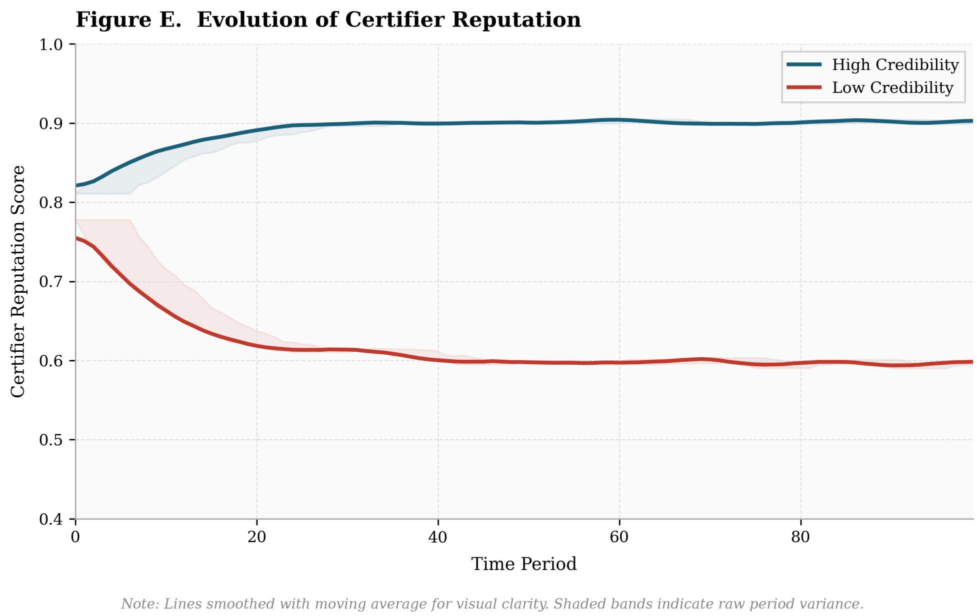

Figure 5: Certifier reputation over time.

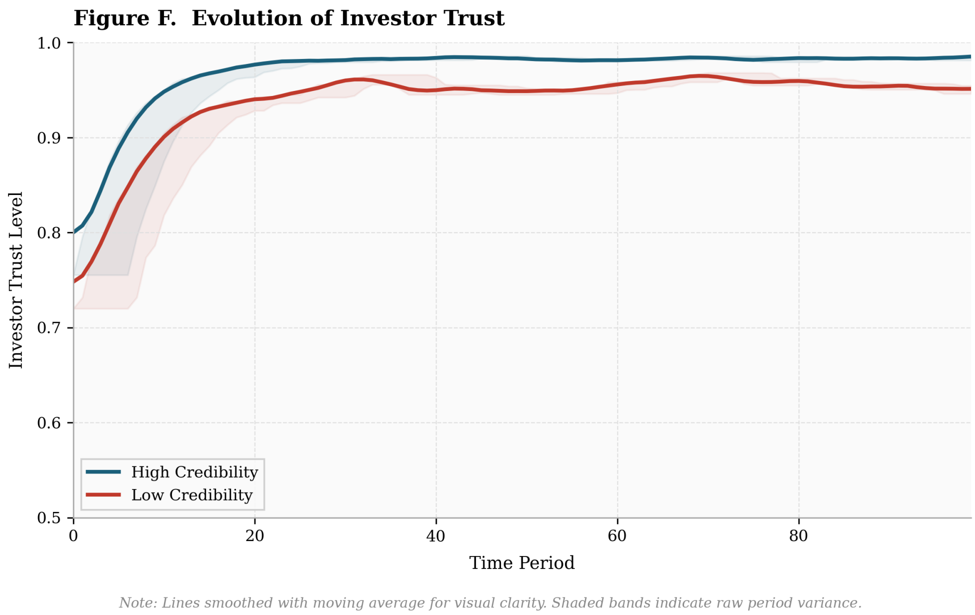

Figure 6: Investor trust over time.

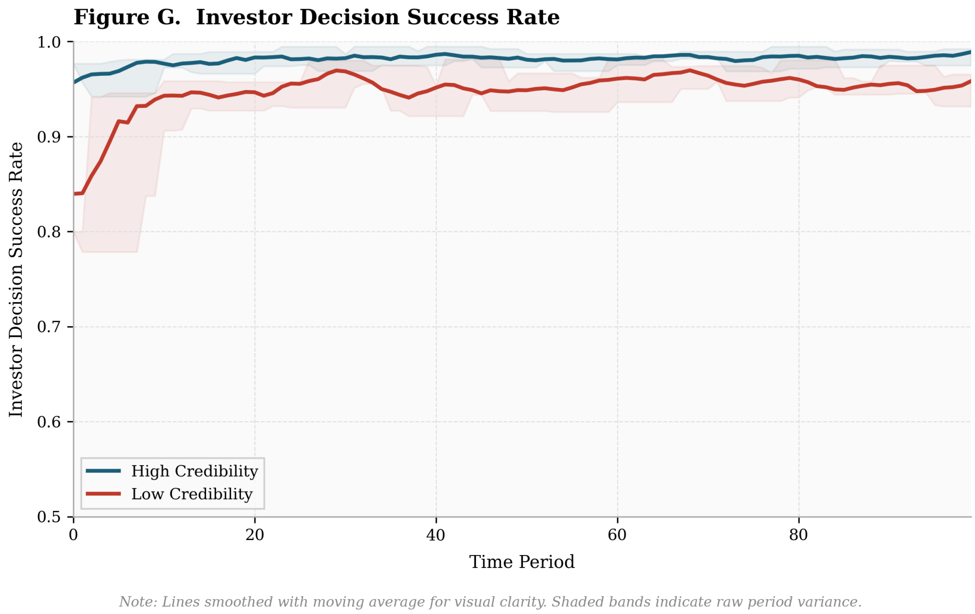

Figure 7: Investor decision success rate over time.